Some tips on preparing and giving a project presentation

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What I will present today

- Advice on how to *prepare*, *write*, and *deliver* a mathematics talk in general, and a project presentation in particular
- What constitutes a good talk depends a lot on your taste, as well as the field, but there are some basics that are universal
- When in doubt, ask your supervisor

What is expected of you

- 20 cp individual projects: Give a 15-minute presentation on your project—this should include 2-3 minutes time for questions
- 40 cp individual projects: Give a 20-minute presentation on your project plus an additional 5 minutes for questions
- 20 cp group projects: Give a 20-minute presentation, divided between group members, plus an additional 5 minutes for questions
- Make it understandable, not too technical
- Can give an overview or concentrate on the most important/interesting part

What is expected of you

Aspect	Comments
Choice of material to talk about (20%)	
Organisation of the talk (20%)	
Quality of the visual aids used (20%)	
Ability to clearly explain concepts verbally (20%)	
Delivery of the talk (pace, volume, timing, audience engagement) (20%)	
General remarks	Talk mark counts for 10% of total mark

Preparing your talk

How to start

- Make sure you know the following:
  - What do you want to convey?
  - Who is your audience? (Experts? Fellow students?)
  - How much time do you have?
  - How will you deliver the talk? (slides, whiteboard, . . .)
  - What is the venue like?
- Make an outline
  - Introduce your talk. What will you tell us? Why should we care?
  - Main body—divide into sections if appropriate. Tell a story!
  - Concluding remarks.

Preparing your talk

Some things to think about:

- Don't include every detail you have learned!
- Simplify. Leave out details that are too technical (you can mention that you do)
- Concentrate on the *story*. What you spent the most time working on is not necessarily what helps the story the most.
  - For pure maths, this might include some background on the history of the problem and why people got interested in it.
  - For applied, talk about how your project (or the general area it lies in) relates to everyday life.
- Think about good presentations/lectures you have seen
- Think about the bad ones too :-)
- How would you explain your project to a friend? How would you explain it to your supervisor?
- If necessary: make at least the first 1/3 of your talk accessible to everyone in the audience, and the rest to experts (but say that this is what you do!)

Writing the talk

- You can use slides (projector) or write on the board. Since the talk is short, using slides of often the safer/better choice.
  - Use *Beamer* to produce typed slides (see the tutorial on Overleaf, or the source for these slides on Blackboard)
  - Handwritten (live or pre-written) slides
- DON'T
  - Do not fill your slides (especially with formulas!)
  - Do not write everything you plan to say
  - Do not use too much notation and specialised terminology
- DO
  - Write enough so the audience can follow
  - Use pictures if appropriate. Hand drawn are perfectly fine!

Actually writing the talk

1. Write a short part of your talk
2. Practise it (and time it)
3. Write the rest
4. Practise it (and time it), preferably in front of someone
5. Proof read, make tweaks
6. Practise, practise, practise!

Writing the talk

- More things to think about:
  - Most common criticism: too technical or went over time
  - You are an expert on the topic, your audience is not!
  - Build in flexibility
    - Have an exit strategy
    - Finish on time! (But not after half the time!)
  - You will probably speak faster during the actual talk

Delivering the talk

- Slow down!
- Look at your audience
- Don't read, don't memorise (practise—but not verbatim)
- Motivate your topic, explain why *you* find it interesting
- Don't assume too much of your audience. Explain your terms, repeat and recall things throughout. (We're bad at paying attention!)
- You are not expected to know everything...
  - ...but don't pretend to know things you don't
  - Instead, if you say things you don't really understand, say so.
  - It is OK to say "I don't understand exactly how, but I have read that this has connections to..." or "My supervisor told me that..."

Delivering the talk

- Answering questions
  - It is OK to pause and think
  - It is OK not to know an answer
    - Just show some interest:
      "Interesting question, I haven't thought about that...";
      "That was outside the scope of my project...";
      "That's something I would definitely like to learn more about..."
    - OK to say that you need to think about it more, ask if you can talk afterwards

Remember

- Have fun!
- Purpose of project talk:
  - 1. Provide nice conclusion to your project where you get to show what you have learned
  - 2. Get some practice in giving presentations
- Worth only 10% of your total mark for the project

Final tips

- Don't say:
  - Consider the—up to multiplicative constant—unique $\mathbb{Z}^2$-invariant locally finite Radon measure on the Cartesian product of two copies of the Hausdorff, locally compact space of real numbers whose topology is generated by intervals, equipped with the product topology
- When you could say:
  - Consider the usual notion of area in the plane $\mathbb{R}^2$
- Don't overuse the `\pause` command in Beamer

Student-led practice session

- Practise your talk with other students and get their feedback
- Scheduled for Wednesday of the week before the presentations:
  - Year 3: 28th January 2026
  - Year 4: 4th March 2026
- To sign up for the session, fill in the form forms.office.com/e/gnWxV44Dkb